MiniOS User Manual

Course: IT3131
Institution: Central Philippine University
Project: MiniOS - Simulated Python Operating System
Group Members: Mark Ellery Contreras, Tamara Nicole M.Dondiego, Shgian Dane Wilfred Delima, Kathy P. Flores, Justenn J. Diel

1. Getting Started
MiniOS is a lightweight, simulated command-line operating system built in Python. Because it runs directly through your computer's terminal, no heavy installation is required.

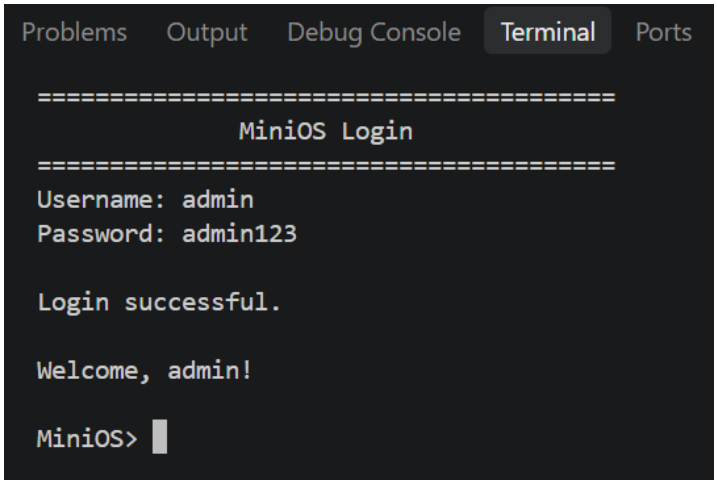
To launch the system:

1. Open your terminal or command prompt.
2. Navigate to the folder containing the MiniOS files.
3. Type the following command and press Enter: `python main.py`

2. Logging In

Security is the first step in MiniOS. Upon launching the system, your screen will clear, and you will be greeted by the secure authentication gateway.

You must enter a valid username and password. The system hides your keystrokes while typing your password to ensure security. Once authenticated, you will be granted access to the `MiniOS>` command prompt



3. Command Guide

MiniOS features 16 built-in commands divided into four core categories. Type any of these commands at the MiniOS> prompt and press Enter.

System & Information Commands

These commands help you navigate the system interface and check your machine's status.

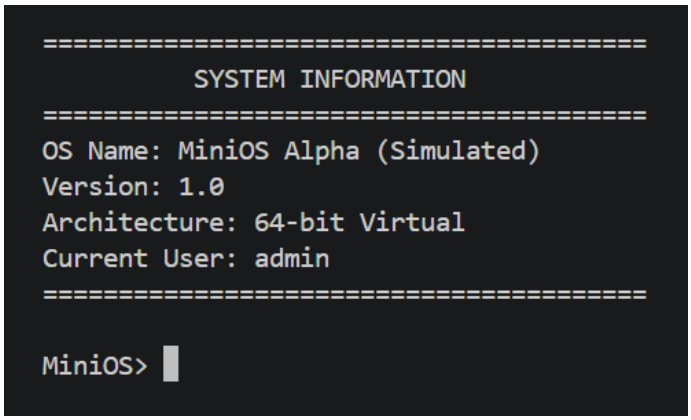
- **help** - Displays the full list of available commands.
- **time** - Shows the current live system date and time.
- **clear** - Wipes the terminal screen clean for easier reading.
- **whoami** - Displays the username of the currently logged-in user.
- **systeminfo** - Shows the OS Name, Version, Architecture, and current user.
- **exit** - Safely shuts down MiniOS and returns you to your normal computer terminal.

File System Management

Manage your simulated files and directories using these commands.

- **dir** - Lists all currently available directories and files.
- **mkdir <name>** - Creates a new folder (e.g., mkdir my_folder).
- **create <name>** - Creates a new empty file (e.g., create notes.txt).
- **read <name>** - Opens and displays the contents of a specific file.

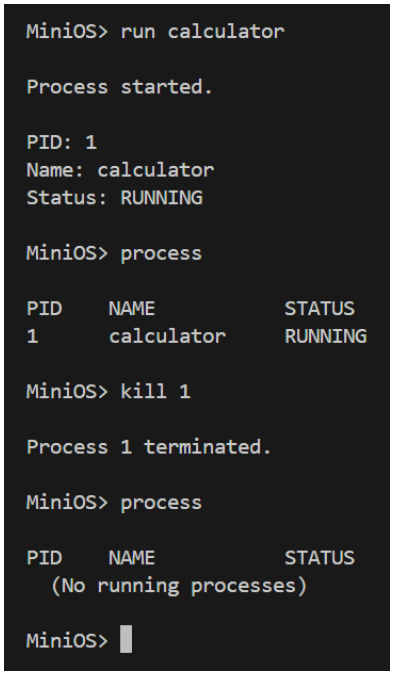
- **delete <name>** - Permanently removes a file from the system.



Process Management

MiniOS allows you to simulate running and terminating background applications.

- **process** - Lists all currently active running processes and their unique PIDs (Process IDs).
- **run <name>** - Starts a new simulated process (e.g., run calculator).
- **kill <pid>** - Terminates a specific running process using its ID number (e.g., kill 1).



Memory Management

Monitor and allocate simulated system RAM.

- **memory** - Displays the total, allocated, and free memory status of the system.
- **allocate <MB>** - Reserves a specific amount of memory in Megabytes (e.g., allocate 50).
- **free <MB>** - Releases a specific amount of previously allocated memory (e.g., free 50).

```
MiniOS> memory

Memory Status
-----------------------
Total Memory: 1024 MB
Used Memory:     0 MB
Free Memory:  1024 MB

MiniOS> allocate 100

100 MB allocated.

MiniOS> memory

Memory Status
-----------------------
Total Memory: 1024 MB
Used Memory:   100 MB
Free Memory:   924 MB

MiniOS> free 100

100 MB released.

MiniOS>
```

Exiting the System

To exit MiniOS, you must use the `exit` command from the `MiniOS>` prompt. Do not attempt to close the terminal window directly while processes are running.

1. Type `exit` and press Enter.
2. The system will gracefully shut down, terminating all background processes.
3. You will be returned to your standard computer command line (e.g., PowerShell or Command Prompt).

